GIACOMO MARCIANI

SENIOR SOFTWARE ENGINEER @ Amazon Web Services

Building High Performance Computing in the Cloud

LOCATION

Boston, MA, United States

CONTACTS

giacomo.marciani@gmail.com
linkedin.com/in/gmarciani
github.com/gmarciani

SUMMARY

I am a Senior Software Engineer with 8+ years at AWS, specializing in High Performance Computing, distributed systems, cloud architectures, infrastructure automation, and technical leadership.

I lead the software development end-to-end: strategy, system design, implementation, CI/CD, observability, security, and on-call operations.

I contributed both to AWS standalone products and services in the HPC field: ParallelCluster, EnginFrame, Elastic Graphics, and Elastic Inference. I worked through every phase of the product lifecycle: beta testing, public launch, growth, maintenance, and sunsetting.

Before AWS, I taught two computer science courses at university, founded the first ACM Student Chapter in Rome, and co-founded a startup. I also led research projects that resulted in scientific publications and presentations at international conferences. In my spare time, I maintain some personal open-source projects.

What makes me a good engineer and leader is simple: I love building software with other people, defining product strategies and helping both customers and developers deliver at their best.

My career goal is to become a Principal Engineer leading mission-critical HPC infrastructures to solve the complex problems that matter most to humankind.

WORK EXPERIENCE

Senior Software Engineer

Amazon Web Services, 2018-Present

· *Senior Software Engineer (2026-Present)*

· *Software Engineer II (2021-2025)*

· *Software Engineer I (2018-2020)*

AWS ParallelCluster (2021-Present). Standalone tool to deploy, operate and monitor HPC clusters on AWS based on NVIDIA Slurm scheduler. I am serving as technical leader of a team of four engineers, owning the software lifecycle end-to-end: strategy, design, implementation, observability, on-call incident response, mentoring, and hiring. My key contributions include multi-user environment, dynamic file system mounting, infrastructure upgrades on running clusters, support for GPU-intensive instance types, enabling the product in air-gapped environments, Terraform integration, ParallelCluster UI, threat modeling, security reviews, driving roadmap and releases, and providing critical support to our most important customers. Recent work includes defining the product strategy for the next two years, introducing a new diagnostic tool for cluster troubleshooting, improving the resilience of in-place cluster infrastructure upgrades, and defining AI agents for the teams to boost productivity in docs, code, and ops.

NICE EnginFrame (2021). Web portal for hybrid HPC clusters. Redefined the build system using Gradle, implemented CI/CD using AWS Code Suite services and worked on the Cloud Connector to interact with both on-prem and cloud clusters.

Elastic Graphics and Elastic Inference (2018-2022, 2025). Services attaching cost-effective GPU acceleration to EC2 instances and ECS containers. Contributed to the launch of Elastic Inference and owned end-to-end the microservice responsible for metering and billing for compute resources. In 2025 I owned the shutdown of those services, coordinating or directly making changes in software owned by 8 different teams and deeply integrated with EC2/ECS.

R&D Intern

Accenture, 2017

Designed a solution for the elastic auto-scaling of containerized applications using Reinforcement Learning in Kubernetes.

Chairman, Founder

ACM Rome Tor Vergata, 2016-2018

Founded the ACM Student Chapter at University of Rome Tor Vergata, the first in the city. Led a team of four master's students and fostered connections between academia and industry. The chapter is still active.

Teaching Assistant

University of Rome Tor Vergata, 2016-2018

Tutored classes of about fifty undergraduate students for a full semester and graded their exams for two courses: Foundations of Computer Science and Algorithms Engineering.

Mathematics Tutor

Freelance, 2011-2017

Gave private math lessons to middle and high school students. All of my students passed their exams, and several of them developed an interest in mathematics that led them to study engineering.

Business Development, Co-Founder

Sbaam.com, e-Commerce, 2011-2012

Co-founded the startup, whose mission was to use optimized QR codes to improve the in-store shopping experience. Responsible for business planning and pitching investors in private meetings and at conferences. The startup was selected as a finalist for the InnovAction Lab startup contest.

SKILLS

High Performance Computing ParallelCluster, Slurm, NVIDIA GPUs, CUDA, MPI, Lustre, EFA, EnginFrame, OSU Benchmarks

Cloud Providers Amazon Web Services, DigitalOcean, Railway

AWS Services EC2, ECS, Lambda, Image Builder, VPC, Route 53, API Gateway, S3, EBS, EFS, FSx, DynamoDB, RDS, CloudFormation, Systems Manager, IAM, KMS, Secrets Manager, Cognito, Directory Service, Inspector, CloudWatch, DCV, CodeBuild, CodeCommit, CodePipeline

Infrastructure as Code CDK, CloudFormation, Terraform, Chef

CI/CD Jenkins, AWS Code Suite

Web Spring, React, Node.js, HTML, Sass, Postman, Hugo

Big Data Flink, Hadoop MapReduce, Pig, Hive, Kafka, Elasticsearch

Languages Python, Java, Kotlin, Ruby, JavaScript, C, CUDA, Bash, MIPS, LaTeX

Data SQL, NoSQL (Document, Key-Value, Graph)

Observability CloudWatch, Grafana, Loki, Prometheus

Security Threat Modeling, Security Reviews

Platforms Linux, macOS, Docker

Applied Math MATLAB, AMPL, IBM CPLEX

Ontology OWL, SPARQL

Leadership Scrum, Roadmap Planning, Project Management, System Design, Mentoring, Hiring

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PROJECTS

AWS ParallelCluster
Amazon Web Services, 2021-Present
Open-source Python tool for infrastructure automation and configuration management of large-scale HPC clusters on AWS. A declarative configuration drives compute, Slurm scheduling, storage, networking, and auto-scaling, with node setup via Chef cookbooks and versioned change rollouts. Drives multi-million-dollar revenue for enterprise and research customers.

AWS ParallelCluster UI
Amazon Web Services, 2022-Present
Open-source web interface to create, monitor, and manage ParallelCluster clusters. Vended as a 1-click CloudFormation stack and implemented as a serverless React application.

NICE EnginFrame
Amazon Web Services, 2021
Web portal to submit and manage HPC workloads across on-premises and AWS clusters, offering a unified interface for job submission, monitoring, and remote 3D visualization.

AWS Elastic Graphics and Elastic Inference
Amazon Web Services, 2018-2022
Services attaching low-cost GPU acceleration for OpenGL and ML inference to EC2, ECS, and SageMaker, cutting the cost of running graphics and machine learning workloads at scale.

Crime Graph
Big Data, IEEE FiCloud, 2017
Data-stream processing application for real-time analysis of evolving criminal networks, applying graph algorithms over Kafka and Flink to surface hidden and potential criminal relations.

CLI Wizard
Developer Tools, 2025
Code generator that produces complete, pip-installable Python CLI projects from OpenAPI v3 specs, with automatic command grouping, profile management, and YAML-based customization.

PyDES
Performance Modeling, 2019
Discrete-event simulation suite for performance modeling of cloud systems, featuring batch-means analysis, Markov-chain solvers, and multi-stream pseudo-random number generation.

OntoQA
Artificial Intelligence, 2017
Question-and-answer web application leveraging ontology-driven natural language processing to map free-text questions onto a knowledge graph and return precise answers.

Godot
Mobility, 2013
Android app that tackles double parking by letting citizens notify the driver through QR/NFC tags that their car must be moved.

See more on my GitHub

PUBLICATIONS

A data streaming approach to link mining in criminal networks
5th IEEE International Conference on Future IoT, 2017

Real-time analysis of social networks leveraging Flink
10th ACM International Conference on Distributed and Event-Based Systems. 2016

Marciani Normal Form of context-free grammars
arXiv, 2016

EDUCATION

MSc Software Engineering and Data Science
University of Rome Tor Vergata, 2015-2018
Grad Avg 29/30 - GPA 4/4 *

BSc Software Engineering
University of Rome Tor Vergata, 2011-2015
Grad 110/110 cum laude

* MSc coursework completed, thesis not submitted. Exams transcript on my LinkedIn

AWARDS

Lecturer at Master CESMA
University of Rome Tor Vergata, 2016

Finalist at ACM DEBS Grand Challenge
10th ACM International Conference on Distributed and Event-Based Systems, 2016

Panelist at InnovAction Lab Grand Challenge
LVenture Groups, 2013

Best Student in Computer Architectures
University of Rome Tor Vergata, 2012

Alumnus of InnovAction Lab
LVenture Groups, 2011

Counselor of Engineering
University of Rome Tor Vergata, 2010

LANGUAGES

Italian
Native

English
Proficient